STORMHACKS 2026 · TEAM HANDOUT

# ATLAS Build Graph

## Who builds what, what can run side by side, what has to wait, and the gates in between.

This is the build plan cut for people. Each box is one area of work that one person owns, about the size of “implement hyperbolic terrain generation”, not a single task. For every phase it shows the areas of each lane, which of them can start at the same time, which one waits for another, what is handed from one person to the next, and the gate each area feeds.

The areas are listed in `build_graph_areas.md`. Everything else on these pages is read from `documents/ATLAS_Master_Plan.md` when the PDF is built: the tasks inside each area, their dependencies and the gates. The tasks themselves, with their outputs and acceptance tests, stay in the plan (§9 and §10) for the coding agents. Work is ordered by dependencies and gates only.

### HOW TO READ AN AREA

- **Starts after** (on the boards: **after**): nothing inside the area can begin until the named area or gate is done.
- **Also needs** (**needs**): the area can start, but one part of it waits for something from another area.
- **Feeds**: the gates at which the area's work is checked.
- **Hands over to**: what another lane is waiting for from this area. These are the moments to talk.
- A tag such as **B·2** is lane B, second area of that phase. Task ids such as `P1–B04` (phase 1, lane B, number 04) only point into the plan. Gates are G0 to G15.

Generated 3 October 2026 from ATLAS_Master_Plan.md (plan last modified 3 October 2026, 17:22).

### LEGEND

🟩 **Lane A** · Platform, Look and Flat Spaces
🟪 **Lane B** · Kernel and Curved Worlds
🟧 **Lane C** · Scan, Room and Safety
🟦 **Lane D** · Link, Station and Forge
⬜ **All hands** · shared work
⬢ **Gate** · where the lanes meet and a check is passed
—— **Solid arrow** · starts after
- - - - **Dashed arrow** · also needs, before it can finish

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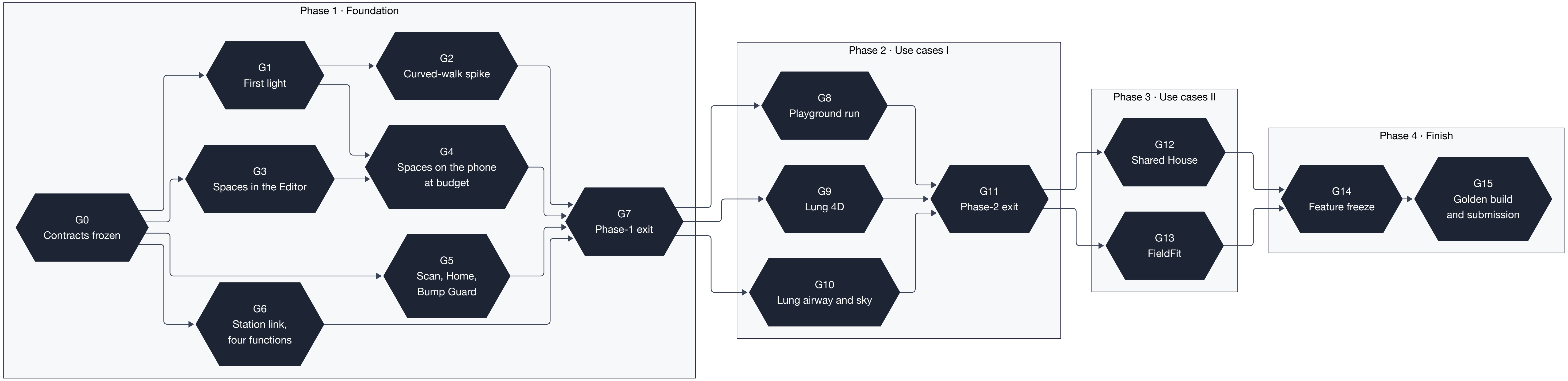
# 1 The big picture

## 1.1 Phases and gates

plan §9.4


Phase 1 · Foundation
G0
Contracts frozen
G1
First light
G2
Curved-walk spike
G3
Spaces in the Editor
G4
Spaces on the phone
at budget
G5
Scan, Home,
Bump Guard
G6
Station link,
four functions
G7
Phase-1 exit
Phase 2 · Use cases I
G8
Playground run
G9
Lung 4D
G10
Lung airway and sky
G11
Phase-2 exit
Phase 3 · Use cases II
G12
Shared House
G13
FieldFit
Phase 4 · Finish
G14
Feature freeze
G15
Golden build
and submission

| Gate | Passes when (summary; full checklists in §11.3) |
|---|---|
| **G0** Contracts frozen | Every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs |
| **G1** First light | Empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build |
| **G2** Curved-walk spike | Five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor |
| **G3** Spaces in the Editor | S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green |
| **G4** Spaces on the phone at budget | Every space S0–S6 meets the Must column cold, with the room's content, on the `budget_room` fixture; the 15-minute soak passes |
| **G5** Scan, Home and Bump Guard | Both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset |
| **G6** Station link, four functions | Mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6 |
| **G7** Phase-1 exit | A first-time walker goes from self-scan to every space, watched and tuned from the Station |
| **G8** Playground run | A first-time walker completes the guided run in two different rooms |
| **G9** Lung 4D | The four 4D pieces run on both case packages at budget |
| **G10** Lung airway and sky | Four forks walked without rescue; sky and plan card correct |
| **G11** Phase-2 exit | Playground and Lung Mission in one session by an outsider, hot, at budget |
| **G12** Shared House | Post Circling walked by three people with no contact |
| **G13** FieldFit | Blind spot found; search is faster with the lens than without |
| **G14** Feature freeze | Full path twice in a row on one phone without cooling; only fixes after this |
| **G15** Golden build and submission | Golden build runs with the Station unplugged; documents, video and metrics delivered |

Phase 3's packages depend technically only on Phase 1. They are placed after G11 because every lane is occupied until then.

## 1.2 The four lanes: who owns what

plan §9.2

| Lane | Phase 1 · Foundation | Phase 2 · Use cases I | Phase 3 · Use cases II | Phase 4 · Finish |
|---|---|---|---|---|
| **A · Platform, Look and Flat Spaces** | Project, build seat, stereo rig, lens blit, shared shader, SpaceLab, Glued (S1), Impossible Rooms (S5), Slicer and `Mesh4D` (S6), room content in Glued, thermal ladder, soak | Slicer toybox, equation sources, look of the Playground; the whole medical 4D runtime (worldtube, breath lane, scanner tilt, tasks) and the Lung director with its labels and credits | Shared House: sheets, door, second home, visitor | Look and audio, performance hardening, golden build |
| **B · Kernel and Curved Worlds** | Kernel, holonomy rig, warp shader, exact tile ids, tile meshes and renderer, world generator, Small Planet (S4), dial and 3-4-5 walk, Aligned Cells (S3), room as tile zero, room content in Aligned Cells | Lung Mission: airway wing in hyperbolic space, valve chambers, Two Compasses, needle sky | FieldFit: lens mathematics, lookup, composed final blit, blind-spot finder | Maths notes, correctness report, equation-card audit, figures |
| **C · Scan, Room and Safety** | AR session, FloorScan core (Editor), scan routes, RoomModel and loop-fit, Home (S0), object photos, Bump Guard with its warning cue and haptics, input, pose pipeline, onboarding, pre-scan path | Geometry Playground: the shell (director, guided walk, captions, twins, menu), the pointing probe, and the Glued and Impossible Rooms exhibits | Shared House: second scan, door fit, Bump Guard on the second sheet, peer link | Onboarding with outside walkers, comfort protocol, failure drills, rehearsals |
| **D · Link, Station and Forge** | `Atlas.Link`, Hub, tuner and Console, perf probe and telemetry, Spectator, flat-twin map, Forge and mesh clean-up, asset client, built-in fallback set, on-device menu, pairing | Lung Mission: the Station pipeline (`atlas case`), case packages, plan card, Lung panels on Console and Spectator | FieldFit: field presets, Console lens panel, Spectator heatmap, search task and logging; Shared House: peer relay (Extended) and two-plan flat twin | Judges' dashboard, session logs, metrics report, video capture, deck |
| Work packages | A 15 · B 12 · C 15 · D 13 (+ 8 shared, 7 gates) | A 17 · B 12 · C 17 · D 13 | A 7 · B 8 · C 5 · D 8 | A 5 · B 5 · C 6 · D 6 |

The lane owner decides inside the lane. Anything that touches a contract (§6.4) is announced to all four before it is committed.

## 1.3 How the plan is organised

plan §9.1

- **By dependency, not by date.** Every task lists what it needs and what proves it is done. A task starts when its inputs exist. A phase ends when its exit gate passes.
- **Contracts first.** The start is wide: three set-up tasks and four sets of contract pages have no dependency, so every engineer and their agents have work from the first minute. They meet in gate G0, where the interfaces between lanes are frozen and stubbed. After that the four lanes do not wait for each other's code until the integration gates.
- **Editor first, phone last.** Only one Mac can build to the phones, so everything that can be proved in the Unity Editor or in a unit test is proved there. The phone is for acceptance.
- **Two levels.** People plan in **work areas**: sixteen in Phase 1, a handful per lane, each a piece one person owns (§9.5.2); later phases have one to three areas per lane and use case (§9.6–9.8). Agents work on **tasks**: each task fits one coding agent, touches one lane's files, and has a binary acceptance test with numbers.
- **The plan's tables and diagrams come from one data set.** The Phase 1 graph has 70 tasks and 166 dependency edges, checked to be free of cycles; the diagrams below are generated from the same data as the tables.

## 1.4 Shared resources and the rules that protect them

plan §9.3

**The build seat (one Mac).**

1. Lane A's first jobs are the build chain (`P1-A02`, `P1-A13`) and the stereo rig (`P1-A03`, `P1-A04`). They take priority over every other lane-A task until G1 passes. Editor-only lane-A tasks (`P1-A06`, `P1-A12`) run alongside them as the graph in §9.5.5 allows.
2. Everyone runs the Unity Editor on their own machine. Native code is wrapped so the Editor and the Station build compile without it.
3. Device builds go out as a build train: one integration build at every gate, plus short on-request slots from a visible queue (who, branch, what is tested).
4. Prefer iteration that needs no rebuild: live tuning from the Console, replay of recorded RoomModels and pose logs in the Editor, kernel and graph logic as EditMode tests.
5. One bundle id and one signing account. A build signed by a different account cannot install over the existing one.

**The two phones.**

1. Phone A is the reference device for every published number. Phone B is lane C's working device (every one of lane C's Phase-1 tasks needs a phone), the hot-swap spare, and the second walker in Phase 3.
2. Hand-held before head-mounted: scan, link and input are debugged with the phone in the hand; the headset is for acceptance.
3. Record once, replay many times: the first good scan and a few walks are saved and replayed in the Editor by every lane.
4. A soak leaves a phone hot. Soaks are scheduled, not squeezed between someone else's measurements, and the thermal state is noted at the start of every measurement.

**The Station's GPU.** The RTX 5070 has 12 GB. The Forge's generator needs about 8 GB and owns the card while jobs are queued. The Spectator runs at modest settings beside it. The medical pipeline (`atlas case`, Phase 2) runs when the Forge is idle.

**Scenes and prefabs.** Unity scenes are single files, so each has exactly one owner.

| Asset | Owner | Rule |
|---|---|---|
| `Atlas_Boot` (XR origin, stereo rig, SpaceLab, bootstraps) | A | Nobody else edits it. Systems register themselves from their own scenes |
| `Space_Home` | C | One additive scene per space, one owner each |
| `Space_Glued`, `Space_Portal`, `Space_Slicer` | A | |
| `Space_CurvedWalk`, `Space_AlignedCells`, `Space_SmallPlanet` | B | |
| `Sys_Scan`, `Sys_BumpGuard`, `Sys_Onboarding` | C | Loaded additively by the boot scene |
| `Sys_Link`, `Sys_Hud`, `PC_Spectator` | D | The Spectator scene is never in the iOS build list |
| Use-case scenes (Phase 2 onwards) | The lane that owns the use case's runtime | Reference spaces through `ISpaceModule`; never copy their objects |
| `Sandbox_<name>` | Each engineer | Never in the build list |
| URP assets, Project Settings, layers, stencil table, package manifest | A | Changed only by A, announced first |
| The shared material and palette | A | Per-tile and per-room material instances are created at run time from a pool, never saved as assets |

**The shared shader.** Ownership of the four include files is in §8.3. A contract check runs at every gate: the shader is still reported SRP-Batcher compatible, and the draw-call count of the stress scene has not grown.

**Source control.** Text serialisation, visible meta files, UnityYAMLMerge as the merge tool, Git LFS for binaries. Small merges to one main branch. Tunables live in the parameter registry and in ScriptableObjects, one file per space, so tuning never touches a scene. Asset Store packs stay out of the repository.

# 2 Phase 1 · Foundation

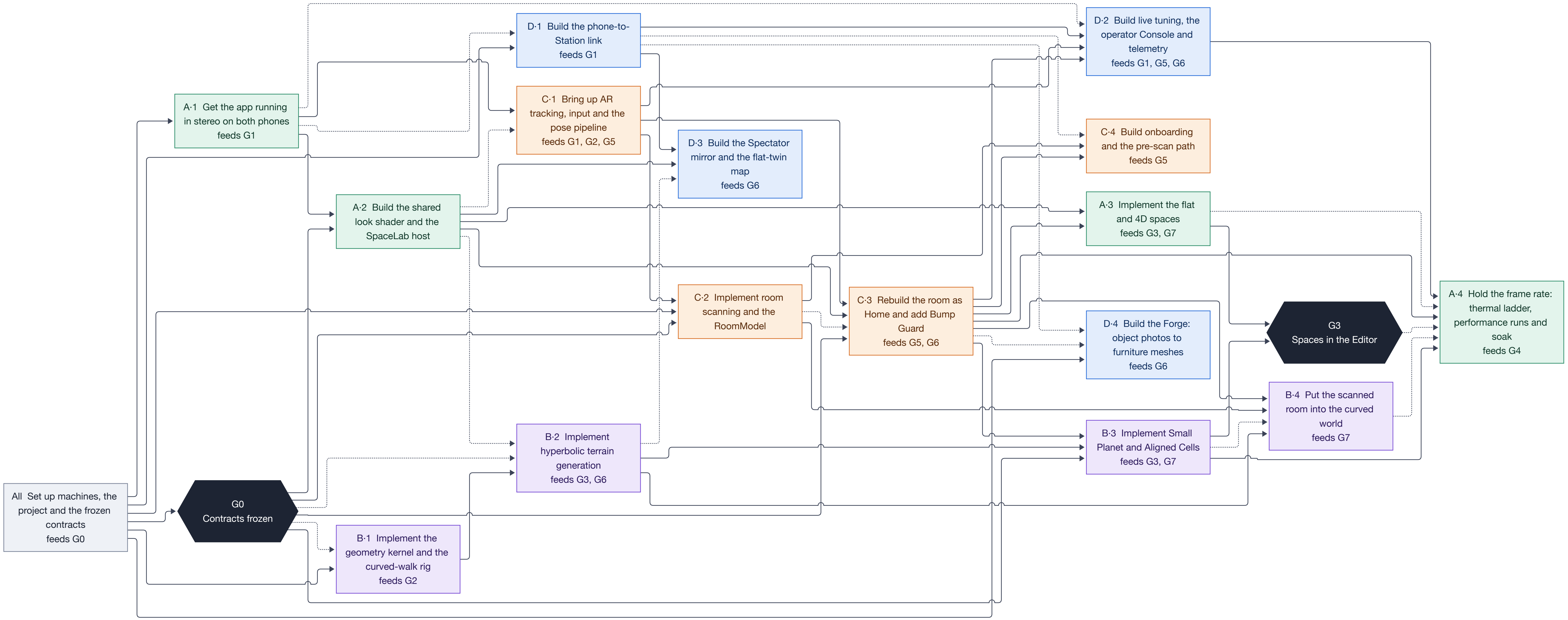
plan §9.5

## 2.1 Who can work side by side

This phase opens the build. 17 areas cover the 63 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person’s order. An area sits in the first step in which part of it can begin.

| Step | Lane A · Platform, Look and Flat Spaces | Lane B · Kernel and Curved Worlds | Lane C · Scan, Room and Safety | Lane D · Link, Station and Forge |
|---|---|---|---|---|
| 1 | **All Set up machines, the project and the frozen contracts**<br>feeds G0 | | | |
| 2 | **A·1 Get the app running in stereo on both phones**<br>after All feeds G1 | **B·1 Implement the geometry kernel and the curved-walk rig**<br>after All needs G0 feeds G2 | **C·2 Implement room scanning and the RoomModel**<br>after All needs G0 C·1 | **D·1 Build the phone-to-Station link**<br>after All needs A·1 feeds G1<br>**D·4 Build the Forge: object photos to furniture meshes**<br>after All needs C·3 D·1 feeds G6 |
| 3 | **A·2 Build the shared look shader and the SpaceLab host**<br>after G0 A·1 | **B·2 Implement hyperbolic terrain generation**<br>after B·1 needs G0 A·2<br>feeds G3 G6 | **C·1 Bring up AR tracking, input and the pose pipeline**<br>after A·1 needs A·2<br>feeds G1 G2 G5 | **D·2 Build live tuning, the operator Console and telemetry**<br>after D·1 needs A·1 C·1 C·3<br>feeds G1 G5 G6 |
| 4 | **A·3 Implement the flat and 4D spaces**<br>after A·2 needs C·3<br>feeds G3 G7 | | **C·3 Rebuild the room as Home and add Bump Guard**<br>after C·1 needs G0 A·2 C·2<br>feeds G5 G6 | **D·3 Build the Spectator mirror and the flat-twin map**<br>after A·2 D·1 needs B·2 feeds G6 |
| 5 | | **B·3 Implement Small Planet and Aligned Cells**<br>after G0 B·2 needs C·3<br>feeds G3 G7 | **C·4 Build onboarding and the pre-scan path**<br>after C·2 needs C·3 D·1 feeds G5 | |
| 6 | **A·4 Hold the frame rate: thermal ladder, performance runs and soak**<br>after C·3 D·2<br>needs G3 A·3 B·3 B·4<br>feeds G4 | **B·4 Put the scanned room into the curved world**<br>after B·2 C·2 needs B·3 C·3<br>feeds G7 | | |

## 2.2 **The area graph**


D·1 Build the phone-to-Station link feeds G1
D·2 Build live tuning, the operator Console and telemetry feeds G1, G5, G6
C·1 Bring up AR tracking, input and the pose pipeline feeds G1, G2, G5
A·1 Get the app running in stereo on both phones feeds G1
D·3 Build the Spectator mirror and the flat-twin map feeds G6
C·4 Build onboarding and the pre-scan path feeds G5
A·2 Build the shared look shader and the SpaceLab host
A·3 Implement the flat and 4D spaces feeds G3, G7
C·2 Implement room scanning and the RoomModel
C·3 Rebuild the room as Home and add Bump Guard feeds G5, G6
D·4 Build the Forge: object photos to furniture meshes feeds G6
G3 Spaces in the Editor
A·4 Hold the frame rate: thermal ladder, performance runs and soak feeds G4
B·4 Put the scanned room into the curved world feeds G7
B·2 Implement hyperbolic terrain generation feeds G3, G6
B·3 Implement Small Planet and Aligned Cells feeds G3, G7
All Set up machines, the project and the frozen contracts feeds G0
G0 Contracts frozen
B·1 Implement the geometry kernel and the curved-walk rig feeds G2

Solid arrow: starts after. Dashed arrow: also needs, before it can finish. Colours are the lanes. A dark hexagon is a gate that areas wait for; the other gates are named in “feeds”. An arrow that is implied by two others is left out.

## 2.3 The areas, lane by lane

### All hands

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **All** **Set up machines, the project and the frozen contracts**<br>every machine and both phones ready, the Station and the network up, the Unity project and repository skeleton, and every contract between the lanes written and stubbed. | nothing in this phase | – | **A·1** Unity project and repository skeleton; Phones and network ready (lane C)<br>**B·1** Unity project and repository skeleton<br>**C·2** Unity project and repository skeleton; Contract pages, lane C<br>**D·1** Contract pages, lane D; Station ready (lane D)<br>**D·4** Station ready (lane D); Contract pages, lane D | **G0** | `9 tasks: P1-A01, X01, X02, X03, X04, X05, X06, X07, X08` |

## Lane A · Platform, Look and Flat Spaces

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **A·1 Get the app running in stereo on both phones**<br>the first build on both phones, URP settings and quality tiers, the two-camera stereo rig, the lens pre-distortion blit, and a one-command build. | All | – | C·1 First build on both phones; Two-camera stereo rig<br>C·3 Two-camera stereo rig<br>D·1 First build on both phones<br>D·2 Two-camera stereo rig | G1 | `5 tasks: P1–A02, A03, A04, A05, A13` |
| **A·2 Build the shared look shader and the SpaceLab host**<br>the one shader and palette that every space uses, and the boot scene that loads one space at a time and runs the frame loop. | G0 A·1 | – | B·2 Shared look shader<br>C·1 SpaceLab host<br>C·3 Shared look shader<br>D·3 Shared look shader; SpaceLab host | – | `2 tasks: P1–A06, A08` |
| **A·3 Implement the flat and 4D spaces**<br>the Glued space (S1) with the scanned room in it, the stencil portal engine and Impossible Rooms (S5), and the Slicer (S6) with Mesh4D. | A·2 | C·3 Home rebuild S0 | – | G3<br>G7 | `5 tasks: P1–A09, A10, A11, A12, A15` |
| **A·4 Hold the frame rate: thermal ladder, performance runs and soak**<br>the thermal quality ladder, a performance run for every space on the budget fixture, and the soak. | C·3 D·2 | G3<br>A·3 Room content in Glued, and standing mode<br>B·3 Room content in Aligned Cells<br>B·4 The room becomes tile zero | – | G4 | `2 tasks: P1–A07, A14` |

## Lane B · Kernel and Curved Worlds

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| B·1 **Implement the geometry kernel and the curved-walk rig**<br>the double-precision kernel for hyperbolic, flat and spherical space, and the holonomy rig that turns head movement into movement through curved space. | All | G0 | – | G2 | `2 tasks: P1–B01, B02` |
| B·2 **Implement hyperbolic terrain generation**<br>the warp shader, exact tile ids and the tile graph, procedural tile meshes, the tile renderer, and the world generator with WorldLint. | B·1 | G0<br>A·2 Shared look shader | D·3 Exact tile ids and the tile graph | G3<br>G6 | `5 tasks: P1–B03, B04, B05, B06, B07` |
| B·3 **Implement Small Planet and Aligned Cells**<br>the spherical space (S4) and the flat-cell mode of the hyperbolic world (S3, Mode 2), with the scanned room in it. | G0 B·2 | C·3 Home rebuild S0 | A·4 Room content in Aligned Cells | G3<br>G7 | `3 tasks: P1–B08, B10, B12` |
| B·4 **Put the scanned room into the curved world**<br>the curvature dial, the 3-4-5 walk, and the scanned room as tile zero of the curved world. | B·2 C·2 | B·3 Small Planet S4<br>C·3 Home rebuild S0 | A·4 The room becomes tile zero | G7 | `2 tasks: P1–B09, B11` |

## Lane C · Scan, Room and Safety

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **C·1 Bring up AR tracking, input and the pose pipeline** the AR session with Editor simulation and replay, gamepad and gaze input with twins, and the pose pipeline. | A·1 | A·2 SpaceLab host | D·2 Input and twins | G1<br>G2<br>G5 | `3 tasks: P1–C01, C10, C11` |
| **C·2 Implement room scanning and the RoomModel** the FloorScan core, the manual and automatic scan routes, object boxes, the RoomPlan wall bridge (Extended), and the RoomModel with the loop-fit. | All | G0<br>C·1 AR session and Editor simulation | B·4 RoomModel and the loop-fit | – | `6 tasks: P1–C02, C03, C04, C05, C06, C15` |
| **C·3 Rebuild the room as Home and add Bump Guard** the Home rebuild (S0), object photo capture for the Forge, the warning cue and haptics, and Bump Guard. | C·1 | G0<br>A·2 Shared look shader<br>C·2 RoomModel and the loop-fit; Object boxes | A·3 Home rebuild S0<br>A·4 Home rebuild S0<br>B·3 Home rebuild S0<br>B·4 Home rebuild S0<br>D·2 Warning cue and haptics<br>D·4 Home rebuild S0 | G5<br>G6 | `4 tasks: P1–C07, C08, C09, C14` |
| **C·4 Build onboarding and the pre-scan path** onboarding with the phone in the hand, and a saved world map that relocalises in a room scanned earlier. | C·2 | C·3 Home rebuild S0<br>D·1 Pairing and network bring-up | – | G5 | `2 tasks: P1–C12, C13` |

## Lane D · Link, Station and Forge

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **D·1** **Build the phone-to-Station link**<br>the Atlas.Link library, the Hub, and pairing and network bring-up. | All | A·1 First build on both phones | C·4 Pairing and network bring-up | G1 | `3 tasks: P1–D01, D02, D11` |
| **D·2** **Build live tuning, the operator Console and telemetry**<br>the parameter registry and Console tuner, the operator Console, the on-device menu, and the perf probe, HUD and telemetry. | D·1 | A·1 Two-camera stereo rig<br>C·1 Input and twins<br>C·3 Warning cue and haptics | A·4 Perf probe, HUD and telemetry; Operator Console | G1<br>G5<br>G6 | `4 tasks: P1–D03, D04, D06, D13` |
| **D·3** **Build the Spectator mirror and the flat-twin map**<br>the Spectator build that re-renders the walker's view on the Station, and the flat-twin mini-map. | A·2 D·1 | B·2 Exact tile ids and the tile graph | – | G6 | `2 tasks: P1–D05, D07` |
| **D·4** **Build the Forge: object photos to furniture meshes**<br>the Forge service on the Station, mesh clean-up, the asset client on the phone, and the built-in fallback set. | All | C·3 Home rebuild S0<br>D·1 Atlas.Link library | – | G6 | `4 tasks: P1–D08, D09, D10, D12` |

## 2.4 Gates of this phase

plan §9.4

| Gate | Needs | Passes when (summary) | Present |
|---|---|---|---|
| G0 **Contracts frozen** | All Set up machines, the project and the frozen contracts | Every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs | all four lanes |
| G1 **First light** | A·1 Get the app running in stereo on both phones<br>C·1 Bring up AR tracking, input and the pose pipeline<br>D·1 Build the phone-to-Station link<br>D·2 Build live tuning, the operator Console and telemetry<br>G0 | Empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build | A, C, D |
| G2 **Curved-walk spike** | B·1 Implement the geometry kernel and the curved-walk rig<br>C·1 Bring up AR tracking, input and the pose pipeline<br>G1 | Five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor | B, C, three walkers from the team, a spotter |
| G3 **Spaces in the Editor** | A·3 Implement the flat and 4D spaces<br>B·2 Implement hyperbolic terrain generation<br>B·3 Implement Small Planet and Aligned Cells<br>G0 | S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green | A, B |
| G4 **Spaces on the phone at budget** | A·4 Hold the frame rate: thermal ladder, performance runs and soak<br>G1 G3 | Every space S0–S6 meets the Must column cold, with the room's content, on the `budget_room` fixture; the 15-minute soak passes | A, B, D |
| G5 **Scan, Home and Bump Guard** | C·1 Bring up AR tracking, input and the pose pipeline<br>C·3 Rebuild the room as Home and add Bump Guard<br>C·4 Build onboarding and the pre-scan path<br>D·2 Build live tuning, the operator Console and telemetry<br>G0 | Both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset | C, B, D |
| G6 **Station link, four functions** | B·2 Implement hyperbolic terrain generation<br>C·3 Rebuild the room as Home and add Bump Guard<br>D·2 Build live tuning, the operator Console and telemetry<br>D·3 Build the Spectator mirror and the flat-twin map<br>D·4 Build the Forge: object photos to furniture meshes<br>G0 | Mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6 | D plus one engineer from each other lane |
| G7 **Phase-1 exit** | A·3 Implement the flat and 4D spaces<br>B·3 Implement Small Planet and Aligned Cells<br>B·4 Put the scanned room into the curved world<br>G2 G4 G5 G6 | A first-time walker goes from self-scan to every space, watched and tuned from the Station | all four lanes |

# 3 Phase 2 · Geometry Playground and Lung Mission

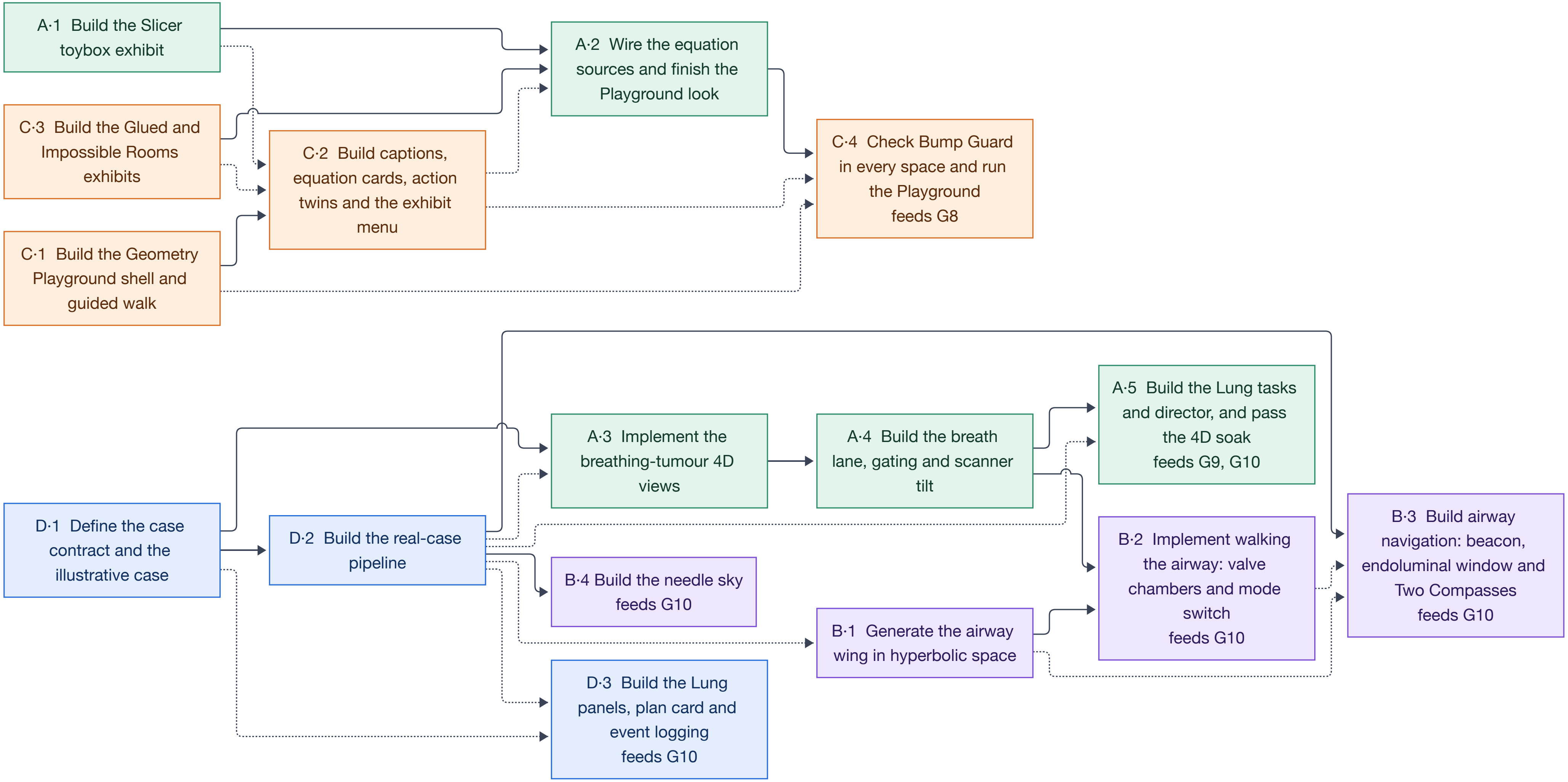
plan §9.6

## 3.1 Who can work side by side

This phase opens when G7 Phase-1 exit has passed. 16 areas cover the 59 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin.

| Step | Lane A · Platform, Look and Flat Spaces | Lane B · Kernel and Curved Worlds | Lane C · Scan, Room and Safety | Lane D · Link, Station and Forge |
|---|---|---|---|---|
| 1 | **A·1 Build the Slicer toybox exhibit** | **B·1 Generate the airway wing in hyperbolic space**<br>needs D·2 | **C·1 Build the Geometry Playground shell and guided walk**<br>**C·3 Build the Glued and Impossible Rooms exhibits** | **D·1 Define the case contract and the illustrative case**<br>**D·3 Build the Lung panels, plan card and event logging**<br>needs D·1 D·2 feeds G10 |
| 2 | **A·2 Wire the equation sources and finish the Playground look**<br>after A·1 C·3 needs C·2<br>**A·3 Implement the breathing-tumour 4D views**<br>after D·1 needs D·2 | | **C·2 Build captions, equation cards, action twins and the exhibit menu**<br>after C·1 needs A·1 C·3 | **D·2 Build the real-case pipeline**<br>after D·1 |
| 3 | **A·4 Build the breath lane, gating and scanner tilt**<br>after A·3 | **B·3 Build airway navigation: beacon, endoluminal window and Two Compasses**<br>after D·2 needs B·1 B·2<br>feeds G10<br>**B·4 Build the needle sky**<br>after D·2 feeds G10 | **C·4 Check Bump Guard in every space and run the Playground**<br>after A·2 needs C·1 C·2 feeds G8 | |
| 4 | **A·5 Build the Lung tasks and director, and pass the 4D soak**<br>after A·4 needs D·2<br>feeds G9 G10 | **B·2 Implement walking the airway: valve chambers and mode switch**<br>after B·1 needs A·4 feeds G10 | | |

## 3.2 The area graph


A·1 Build the Slicer toybox exhibit
A·2 Wire the equation sources and finish the Playground look
C·3 Build the Glued and Impossible Rooms exhibits
C·2 Build captions, equation cards, action twins and the exhibit menu
C·4 Check Bump Guard in every space and run the Playground feeds G8
C·1 Build the Geometry Playground shell and guided walk
A·5 Build the Lung tasks and director, and pass the 4D soak feeds G9, G10
A·3 Implement the breathing-tumour 4D views
A·4 Build the breath lane, gating and scanner tilt
B·3 Build airway navigation: beacon, endoluminal window and Two Compasses feeds G10
D·1 Define the case contract and the illustrative case
D·2 Build the real-case pipeline
B·2 Implement walking the airway: valve chambers and mode switch feeds G10
B·4 Build the needle sky feeds G10
B·1 Generate the airway wing in hyperbolic space
D·3 Build the Lung panels, plan card and event logging feeds G10

Solid arrow: starts after. Dashed arrow: also needs, before it can finish. Colours are the lanes. A dark hexagon is a gate that areas wait for; the other gates are named in “feeds”. An arrow that is implied by two others is left out.

## 3.3 The areas, lane by lane

### Lane A · Platform, Look and Flat Spaces

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **A·1 Build the Slicer toybox exhibit**<br>the Slicer toybox with section names and shadow. | nothing in this phase | – | C·2 Slicer toybox | – | `2 tasks: P2-A01, A02` |
| **A·2 Wire the equation sources and finish the Playground look**<br>the live numbers behind the equation cards of the exhibits, and the look polish of the Playground. | A·1 C·3 | C·2 Caption chips and equation cards | C·4 Look polish; Exhibit equation sources | – | `2 tasks: P2-A03, A04` |
| **A·3 Implement the breathing-tumour 4D views**<br>the 4D tube loaded into Mesh4D, live slices with phase colouring, shells and the look-down shadow, the breathing chest, and the ring fallback. | D·1 | D·2 atlas case chest | – | – | `6 tasks: P2-A20, A21, A22, A23, A24, A30` |
| **A·4 Build the breath lane, gating and scanner tilt**<br>the breath lane with the unroll animation and "now" plane, the gating flags, and the scanner tilt. | A·3 | – | B·2 Unroll animation and "now" plane | – | `4 tasks: P2-A25, A26, A27, A28` |
| **A·5 Build the Lung tasks and director, and pass the 4D soak**<br>tasks M2, M3 and M4 with scoring and events, the Lung director with labels and credits, and the Lung 4D soak report. | A·4 | D·2 atlas case pack and delivery | – | G9<br>G10 | `3 tasks: P2-A29, A31, A32` |

## Lane B · Kernel and Curved Worlds

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| **B·1** **Generate the airway wing in hyperbolic space**<br>the airway tree mapped onto the tile graph, and the corridor and chamber kit. | nothing in this phase | **D·2** atlas case airway | – | – | `2 tasks: P2–B01, B02` |
| **B·2** **Implement walking the airway: valve chambers and mode switch**<br>the valve solver for Mode 1, the valve walk for Mode 2, the mode switch, and the lesion chamber hand-over. | **B·1** | **A·4** Unroll animation and "now" plane | – | **G10** | `4 tasks: P2–B03, B04, B05, B09` |
| **B·3** **Build airway navigation: beacon, endoluminal window and Two Compasses**<br>the beacon and route cost, the endoluminal window, Two Compasses, and the airway run test. | **D·2** | **B·1** AirwayTree<br>**B·2** Mode switch | – | **G10** | `4 tasks: P2–B06, B07, B08, B12` |
| **B·4** **Build the needle sky**<br>the needle sky with its phase and robust layers, and the great-circle check. | **D·2** | – | – | **G10** | `2 tasks: P2–B10, B11` |

## Lane C · Scan, Room and Safety

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| C·1 **Build the Geometry Playground shell and guided walk**<br>the director and beats, scan-to-Home, the centre mark and dial hand-off, the guided 3-4-5 walk, the mode compare, tutorial prompts with the comfort question, and the pointing probe. | nothing in this phase | – | – | – | `7 tasks: P2–C01, C02, C03, C04, C05, C09, C17` |
| C·2 **Build captions, equation cards, action twins and the exhibit menu**<br>caption chips and equation cards, the action twins, and the exhibit menu with return to Home. | C·1 | A·1 Slicer toybox<br>C·3 Glued exhibit: wrap-walk; Impossible Rooms exhibit | A·2 Caption chips and equation cards | – | `3 tasks: P2–C06, C07, C08` |
| C·3 **Build the Glued and Impossible Rooms exhibits**<br>the Glued exhibit with the chiral kit and the gluing switch, and the Impossible Rooms exhibit with the loop corridor. | nothing in this phase | – | A·2 Glued exhibit: wrap-walk; Impossible Rooms exhibit; Chiral kit; Gluing switch and $T^3$; Loop corridor and room numbers | – | `5 tasks: P2–C12, C13, C14, C15, C16` |
| C·4 **Check Bump Guard in every space and run the Playground**<br>the Bump Guard check repeated in every space, and the complete Playground run. | A·2 | C·1 Scan-to-Home; Mode compare; Tutorial prompts and FMS; Pointing probe and SEQ<br>C·2 Action twins; Exhibit menu and return Home | – | G8 | `2 tasks: P2–C10, C11` |

## Lane D · Link, Station and Forge

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| D·1 **Define the case contract and the illustrative case**<br>the case contract with its readers, and the illustrative case package that lanes A and B build against. | nothing in this phase | – | A·3 Case contract; atlas case twin | – | `2 tasks: P2–D01, D02` |
| D·2 **Build the real-case pipeline**<br>the `atlas case` steps from fetch and masks to the 4D tube, shells, chest, airway and sky, packed and delivered to the phone. | D·1 | – | A·3 atlas case chest<br>A·5 atlas case pack and delivery<br>B·1 atlas case airway<br>B·3 atlas case airway<br>B·4 atlas case sky | – | `7 tasks: P2–D03, D04, D05, D06, D07, D08, D09` |
| D·3 **Build the Lung panels, plan card and event logging**<br>the Lung Console panel, the Lung twins on Spectator and Console, the plan card, and event logging with the pilot export. | nothing in this phase | D·1 Case contract<br>D·2 atlas case airway | – | G10 | `4 tasks: P2–D10, D11, D12, D13` |

## 3.4 Gates of this phase

plan §9.4

| Gate | Needs | Passes when (summary) | Present |
|---|---|---|---|
| G8 **Playground run** | C·4 Check Bump Guard in every space and run the Playground<br>G7 | A first-time walker completes the guided run in two different rooms | A, C, two first-time users |
| G9 **Lung 4D** | A·5 Build the Lung tasks and director, and pass the 4D soak<br>G7 | The four 4D pieces run on both case packages at budget | A, B, D |
| G10 **Lung airway and sky** | A·5 Build the Lung tasks and director, and pass the 4D soak<br>B·2 Implement walking the airway: valve chambers and mode switch<br>B·3 Build airway navigation: beacon, endoluminal window and Two Compasses<br>B·4 Build the needle sky<br>D·3 Build the Lung panels, plan card and event logging<br>G7 | Four forks walked without rescue; sky and plan card correct | B, D |
| G11 **Phase-2 exit** | G8 G9 G10 | Playground and Lung Mission in one session by an outsider, hot, at budget | all four lanes |

## 3.5 What the plan says about this phase

plan §9.6

> Two tracks run in parallel from G7. The specifications and the work-package tables are in §10.1 (Playground) and §10.2 (Lung Mission); the diagrams here show the order.
>
> | Track | Lanes | Packages | Exit gate |
> |---|---|---|---|
> | Geometry Playground | C (shell; Glued and Impossible Rooms exhibits) and A (Slicer toybox, equation sources, look) | `P2-C01`–`P2-C17`, `P2-A01`–`P2-A04` | G8 |
> | Lung Mission | D (Station pipeline, case packages, plan card), A (medical 4D runtime), B (airway wing, needle sky) | `P2-D01`–`P2-D13`, `P2-A20`–`P2-A32`, `P2-B01`–`P2-B12` | G9 (4D), G10 (airway and sky) |
>
> Lane A serves both tracks, which departs from a strict "two lanes each" (open item 17 in §14): the Slicer toybox and the medical 4D runtime both build on lane A's `Mesh4D`. Lane A does `P2-A01` and `P2-A02` first, because `P2-A01` unblocks lane C's exhibit menu (`P2-C08`), then starts `P2-A20` onwards; `P2-A03` and `P2-A04` follow when their lane-C inputs land.
>
> What is sequential:
>
> - In the Playground everything hangs off the director (`P2-C01`); the exhibits wrap Phase 1 spaces and are independent of one another; the run (`P2-C11`) is last.
> - In the Lung Mission the case contract (`P2-D01`) comes first. Lane D then produces the **illustrative** package (`P2-D02`) before the real one, so lanes A and B have data to build against without waiting for the full pipeline. The 4D runtime (lane A) and the airway wing (lane B) are independent of each other until the lesion chamber hand-over (`P2-B09`). The Lung director (`P2-A32`) stubs the beats that belong to lanes B and D, so G9 does not wait for the airway.
> - G11 needs G8, G9 and G10, and then one combined session by someone from outside the team.
>
> **Work areas, Phase 2**
>
> The areas are for people; the packages inside them are for agents. An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.

# 4 Phase 3 · Shared House and FieldFit

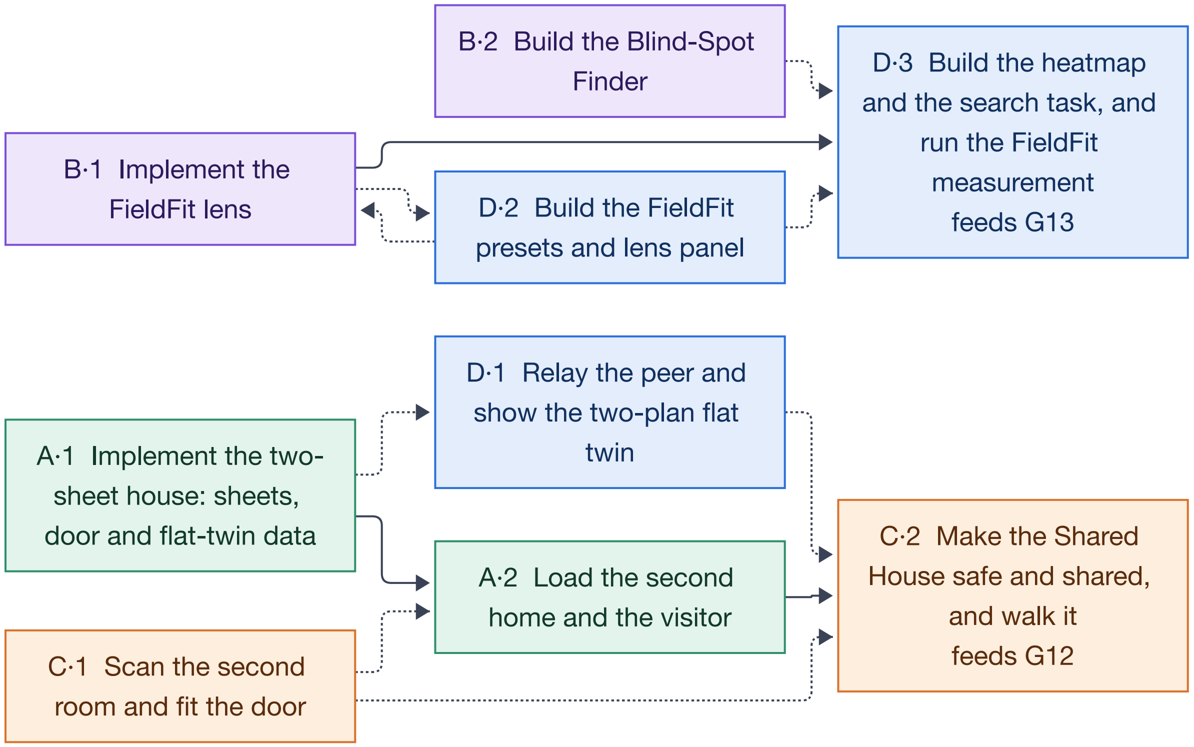
plan §9.7

## 4.1 Who can work side by side

This phase opens when **G11** Phase-2 exit has passed. 9 areas cover the 28 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person’s order. An area sits in the first step in which part of it can begin.

| Step | Lane A · Platform, Look and Flat Spaces | Lane B · Kernel and Curved Worlds | Lane C · Scan, Room and Safety | Lane D · Link, Station and Forge |
|---|---|---|---|---|
| 1 | **A·1 Implement the two-sheet house: sheets, door and flat-twin data** | **B·1 Implement the FieldFit lens**<br>needs D·2<br>**B·2 Build the Blind-Spot Finder** | **C·1 Scan the second room and fit the door** | **D·1 Relay the peer and show the two-plan flat twin**<br>needs A·1<br>**D·2 Build the FieldFit presets and lens panel**<br>needs B·1 |
| 2 | **A·2 Load the second home and the visitor**<br>after A·1 needs C·1 | | | **D·3 Build the heatmap and the search task, and run the FieldFit measurement**<br>after B·1 needs B·2 D·2<br>feeds G13 |
| 3 | | | **C·2 Make the Shared House safe and shared, and walk it**<br>after A·2 needs C·1 D·1<br>feeds G12 | |

## 4.2 The area graph


B·2 Build the Blind-Spot Finder
D·3 Build the heatmap and the search task, and run the FieldFit measurement feeds G13
B·1 Implement the FieldFit lens
D·2 Build the FieldFit presets and lens panel
D·1 Relay the peer and show the two-plan flat twin
A·1 Implement the two-sheet house: sheets, door and flat-twin data
C·2 Make the Shared House safe and shared, and walk it feeds G12
A·2 Load the second home and the visitor
C·1 Scan the second room and fit the door

Solid arrow: starts after. Dashed arrow: also needs, before it can finish. Colours are the lanes. A dark hexagon is a gate that areas wait for; the other gates are named in “feeds”. An arrow that is implied by two others is left out.

## 4.3 The areas, lane by lane

### Lane A · Platform, Look and Flat Spaces

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| A·1 **Implement the two-sheet house: sheets, door and flat-twin data**<br>the sheet state machine, the two-sheet door rendering, and the winding counter with the flat-twin data. | nothing in this phase | – | D·1 Winding counter and flat-twin data | – | `3 tasks: P3-A01, A02, A05` |
| A·2 **Load the second home and the visitor**<br>the second home loader, the visitor avatar with its recorded track, the phone performance pass, and the guest rug if the lead approves it. | A·1 | C·1 Door placement and door fit | C·2 Second home loader; Visitor avatar, recorded track, labels; Phone performance pass | – | `4 tasks: P3-A03, A04, A06, A07` |

### Lane B · Kernel and Curved Worlds

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| B·1 **Implement the FieldFit lens**<br>the lens mathematics, the lookup builder, the wide source render with the composed final blit, the field-loss mask and slider, the lens guard, and the phone performance pass. | nothing in this phase | D·2 Field presets | D·2 Lookup builder<br>D·3 Lens maths library; Field-loss mask and slider; Phone performance pass | – | `6 tasks: P3-B01, B02, B03, B04, B05, B08` |
| B·2 **Build the Blind-Spot Finder**<br>the Blind-Spot Finder, and Scotoma River (Extended). | nothing in this phase | – | D·3 Blind-Spot Finder | – | `2 tasks: P3-B06, B07` |

### Lane C · Scan, Room and Safety

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| C·1 **Scan the second room and fit the door**<br>the second scan path, and door placement with the door fit. | nothing in this phase | – | A·2 Door placement and door fit | – | `2 tasks: P3-C01, C02` |
| C·2 **Make the Shared House safe and shared, and walk it**<br>Bump Guard on the second sheet, the peer link (Extended), and the G12 walk-through. | A·2 | C·1 Door placement and door fit<br>D·1 Peer relay (Extended); Two-plan flat twin | – | G12 | `3 tasks: P3-C03, C04, C05` |

### Lane D · Link, Station and Forge

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| D·1 **Relay the peer and show the two-plan flat twin**<br>the peer relay (Extended) and the two floor plans with the winding dial on the Station. | nothing in this phase | A·1 Winding counter and flat-twin data | C·2 Peer relay (Extended); Two-plan flat twin | – | `2 tasks: P3–D07, D08` |
| D·2 **Build the FieldFit presets and lens panel**<br>the field presets, the Console lens panel, and the Teichmüller lookup (Stretch). | nothing in this phase | B·1 Lookup builder | B·1 Field presets | – | `3 tasks: P3–D01, D02, D05` |
| D·3 **Build the heatmap and the search task, and run the FieldFit measurement**<br>the Spectator heatmap and bound, the search task with logging, and the G13 measurement run. | B·1 | B·2 Blind-Spot Finder<br>D·2 Console lens panel | – | G13 | `3 tasks: P3–D03, D04, D06` |

## 4.4 Gates of this phase

plan §9.4

| Gate | Needs | Passes when (summary) | Present |
|---|---|---|---|
| G12 **Shared House** | C·2 Make the Shared House safe and shared, and walk it<br>G11 | Post Circling walked by three people with no contact | A, C, both phones |
| G13 **FieldFit** | D·3 Build the heatmap and the search task, and run the FieldFit measurement<br>G11 | Blind spot found; search is faster with the lens than without | B, D |

## 4.5 What the plan says about this phase

plan §9.7

> Two independent use cases, two lanes each; lane D adds two small packages to the Shared House (the peer relay and the two-plan flat twin). Specifications and work-package tables: §10.3 and §10.4. Packages marked Extended, Stretch or Blocked are drawn but feed no gate.
>
> **Work areas, Phase 3**
>
> An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.

# 5 Phase 4 · Finish

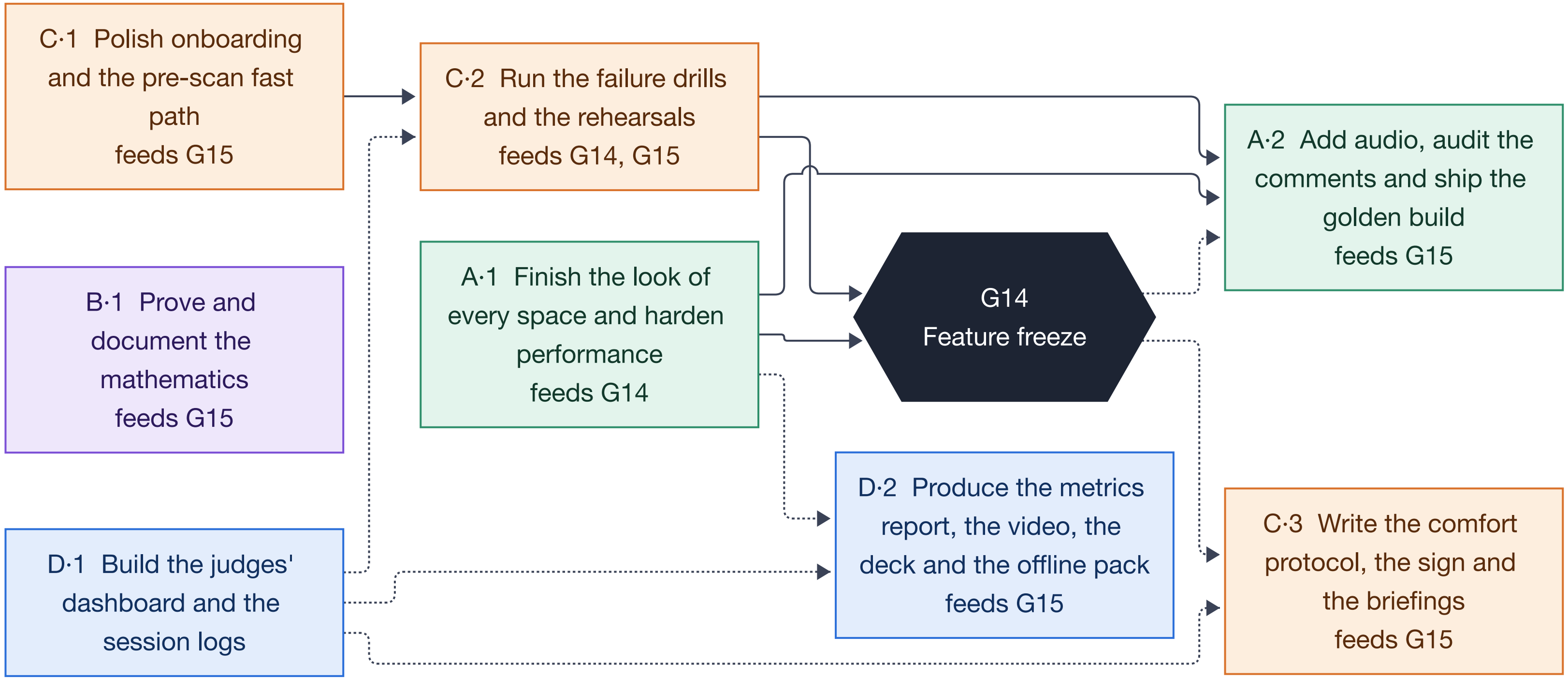
plan §9.8

## 5.1 Who can work side by side

This phase opens when G12 Shared House and G13 FieldFit have passed. 8 areas cover the 22 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin.

| Step | Lane A · Platform, Look and Flat Spaces | Lane B · Kernel and Curved Worlds | Lane C · Scan, Room and Safety | Lane D · Link, Station and Forge |
|---|---|---|---|---|
| 1 | **A·1 Finish the look of every space and harden performance**<br>feeds G14 | **B·1 Prove and document the mathematics**<br>feeds G15 | **C·1 Polish onboarding and the pre-scan fast path**<br>feeds G15 | **D·1 Build the judges' dashboard and the session logs**<br>**D·2 Produce the metrics report, the video, the deck and the offline pack**<br>needs A·1 D·1 feeds G15 |
| 2 | | | **C·2 Run the failure drills and the rehearsals**<br>after C·1 needs D·1<br>feeds G14 G15<br>**C·3 Write the comfort protocol, the sign and the briefings**<br>needs G14 D·1 feeds G15 | |
| 3 | **A·2 Add audio, audit the comments and ship the golden build**<br>after A·1 C·2 needs G14<br>feeds G15 | | | |

## 5.2 The area graph


C·1 Polish onboarding and the pre-scan fast path
feeds G15
C·2 Run the failure drills and the rehearsals
feeds G14, G15
A·2 Add audio, audit the comments and ship the golden build
feeds G15
B·1 Prove and document the mathematics
feeds G15
A·1 Finish the look of every space and harden performance
feeds G14
G14
Feature freeze
D·2 Produce the metrics report, the video, the deck and the offline pack
feeds G15
C·3 Write the comfort protocol, the sign and the briefings
feeds G15
D·1 Build the judges' dashboard and the session logs

Solid arrow: starts after. Dashed arrow: also needs, before it can finish. Colours are the lanes. A dark hexagon is a gate that areas wait for; the other gates are named in “feeds”. An arrow that is implied by two others is left out.

## 5.3 The areas, lane by lane

### Lane A · Platform, Look and Flat Spaces

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| A·1 **Finish the look of every space and harden performance**<br>the look pass for every space and the soak on both phones, which is what the feature freeze waits for. | nothing in this phase | – | D·2 Performance hardening | G14 | `2 tasks: P4–A01, A03` |
| A·2 **Add audio, audit the comments and ship the golden build**<br>the audio loops and cues, the comment audit, and the golden build with its launch check. | A·1 C·2 | G14 | – | G15 | `3 tasks: P4–A02, A04, A05` |

### Lane B · Kernel and Curved Worlds

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| B·1 **Prove and document the mathematics**<br>the maths notes, the correctness report, the equation-card audit, the figure data, and the prior-art note. | nothing in this phase | – | – | G15 | `5 tasks: P4–B01, B02, B03, B04, B05` |

### Lane C · Scan, Room and Safety

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| C·1 **Polish onboarding and the pre-scan fast path**<br>onboarding polished with outside walkers, and the pre-scan fast-path drill. | nothing in this phase | – | – | G15 | `2 tasks: P4–C01, C03` |
| C·2 **Run the failure drills and the rehearsals**<br>the failure drills and the rehearsals with outside walkers. | C·1 | D·1 Judges' dashboard | A·2 Failure drills | G14<br>G15 | `2 tasks: P4–C04, C05` |
| C·3 **Write the comfort protocol, the sign and the briefings**<br>the comfort protocol, the booth sign and the spotter briefing. | nothing in this phase | G14<br>D·1 Session log and pilot data | – | G15 | `2 tasks: P4–C02, C06` |

### Lane D · Link, Station and Forge

| Area and what it delivers | Starts after | Also needs, before it can finish | Hands over to other lanes | Feeds gate | Inside, in the plan |
|---|---|---|---|---|---|
| D·1 **Build the judges' dashboard and the session logs**<br>the judges' dashboard, and the session log with the pilot data. | nothing in this phase | – | C·2 Judges' dashboard<br>C·3 Session log and pilot data | – | `2 tasks: P4-D01, D02` |
| D·2 **Produce the metrics report, the video, the deck and the offline pack**<br>the metrics report, the Spectator capture, the deck with its claims check, and the offline pack. | nothing in this phase | A·1 Performance hardening<br>D·1 Session log and pilot data; Judges' dashboard | – | G15 | `4 tasks: P4-D03, D04, D05, D06` |

## 5.4 Gates of this phase

plan §9.4

| Gate | Needs | Passes when (summary) | Present |
|---|---|---|---|
| G14 **Feature freeze** | A·1 Finish the look of every space and harden performance<br>C·2 Run the failure drills and the rehearsals<br>G12 G13 | Full path twice in a row on one phone without cooling; only fixes after this | all four lanes |
| G15 **Golden build and submission** | A·2 Add audio, audit the comments and ship the golden build<br>B·1 Prove and document the mathematics<br>C·1 Polish onboarding and the pre-scan fast path<br>C·2 Run the failure drills and the rehearsals<br>C·3 Write the comfort protocol, the sign and the briefings<br>D·2 Produce the metrics report, the video, the deck and the offline pack<br>G14 | Golden build runs with the Station unplugged; documents, video and metrics delivered | all four lanes |

## 5.5 What the plan says about this phase

plan §9.8

> Polish, evaluation, demo operations and documentation. Work-package table: §13.6. Feature freeze (G14) follows the look pass, the performance hardening and the failure drills; after it, only fixes, audio, documents and rehearsals. In the diagrams of §9.7 and §9.8 the start gate shows when a lane is free; the tables give the technical dependency (§9.4).
>
> **Work areas, Phase 4**
>
> An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.

# 6 Working with AI coding agents

plan §9.9

The tasks are sized and worded to be handed over. The rules that make that safe:

1. **One task, one branch, one merge request.** The branch name is the task id.
2. **An agent stays in its lane's folders.** It may read contracts and other lanes' public interfaces; it may not edit them. A contract change is made by a human and announced.
3. **The acceptance test comes first.** For anything provable in the Editor the agent writes the test from the acceptance cell, sees it fail, then implements.
4. **Parallel agents take tasks marked ∥** in §9.5.1: they touch different files by construction.
5. **A human runs every phone test** and records the numbers. An agent prepares the build, the test scene and the checklist.
6. **No agent merges on a red test or a broken shader contract.**
7. **Numbers in a task are part of the task.** If a budget cannot be met, the agent reports the measured value and stops; it does not relax the number.

A task brief, ready to paste:

```
Task        P1-B04  Exact tile ids and the tile graph            Lane B
Read first  docs/contracts/kernel.md; this plan §3.4, §6.5, Appendix A.5
Depends on  P1-B01 (merged)
You own     Assets/Atlas/Kernel/Tiling/**  and its tests. Touch nothing else.
Deliver     TileId.cs, TileGraph.cs, Tests/TileIdTests.cs
Accept      - per-depth tile counts equal OEIS A377322 to depth 10
            - a walk round a vertex closes after exactly 5 steps
            - φ(X²+Y²) - T² = -1 for every generated tile (evaluated in Int128)
            - tile centres within ρ = 2 / 3 / 3.5 / 4 of a tile centre: 13 / 45 / 81 / 129; within 3.8425: 105
            - a tile crossing costs ≤ 0.5 ms in the Editor
Out of scope  rendering, materials, scenes, shader code
Report      test log; measured crossing cost; anything in the contract you could not satisfy
```